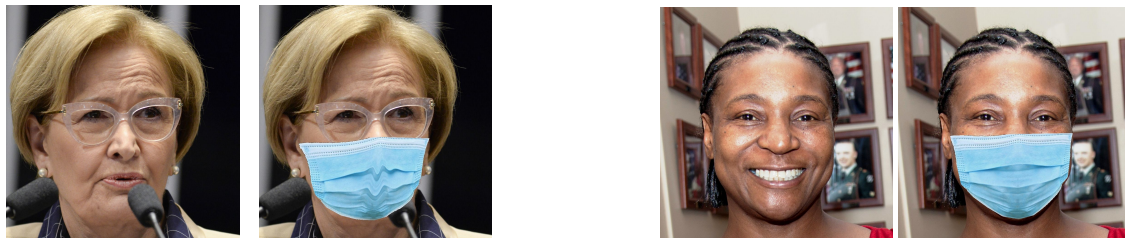


Face-Mask Generation with Stable Diffusion Inpainting

Hok Seng, Mohamed Amine Amrani

Realistic masked-face synthesis can expand limited real datasets with scalable training examples

- Real masked-face datasets are limited, while collecting large facial datasets is costly and privacy-sensitive
- Synthetic generation can increase identity and mask-style diversity while keeping the data creation process controllable
- The aim of this project is to generate a realistic masked face from a clean portrait and compare diffusion against a GAN baseline



Bibliography and research setting

Reference literature

- S. Mumford proposed a GAN-based masked-face generation framework and reported **CycleGAN** as the strongest baseline [1]
- **Latent Diffusion Models** [2] introduced powerful pretrained image priors, making diffusion a strong alternative to GAN-based image translation
- **LoRA** allows adapting large pretrained models by training only low-rank weight updates, significantly reducing computational cost while preserving the pretrained representation [3]
- This project investigates two diffusion-based frameworks:
 - a conditional diffusion model trained from scratch
 - a Stable Diffusion inpainting model fine-tuned with LoRA

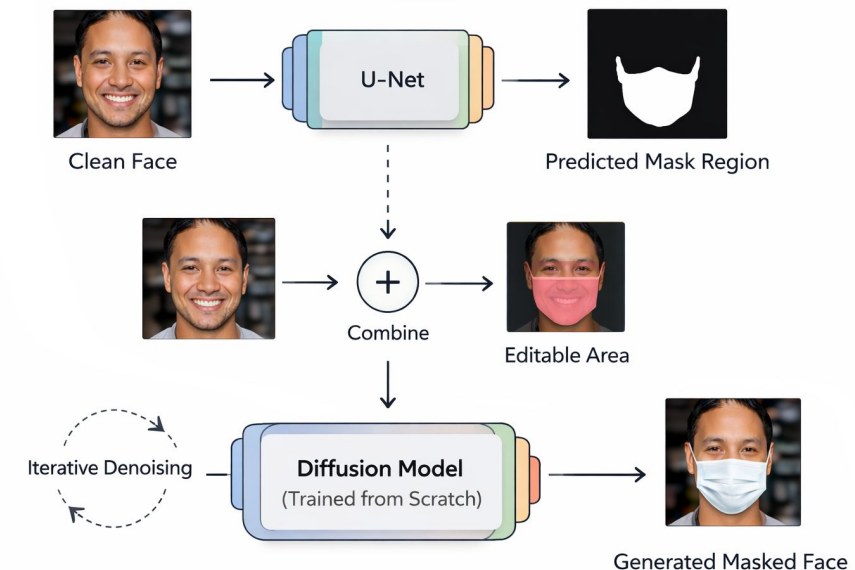
Benchmark setting

For training and test we used aligned face images from **FFHQ** [4] as the clean source domain and **MaskedFace-Net (CMFD)** [5] as the masked target domain.

Proposed methodology

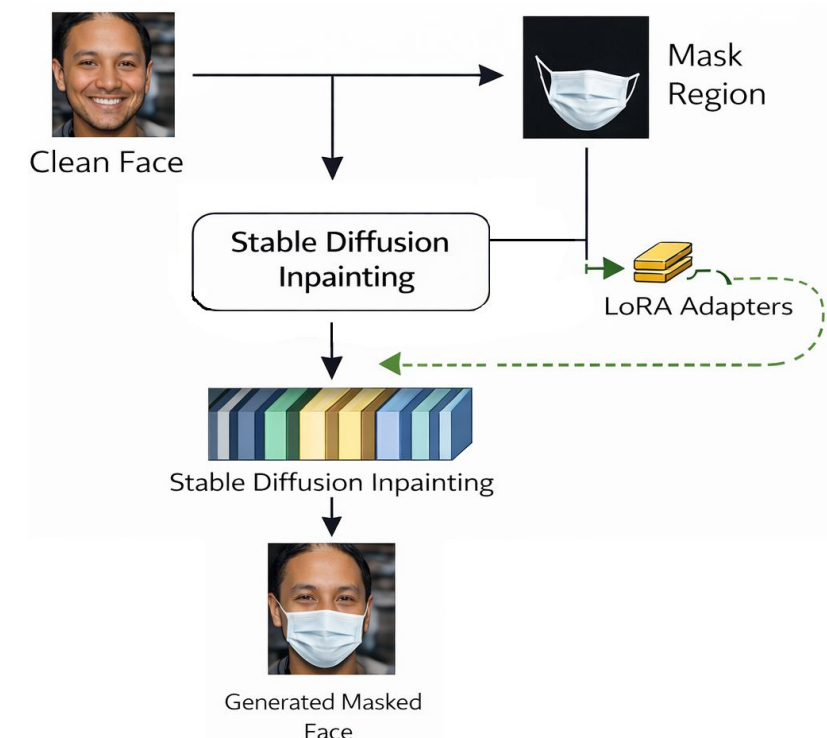
DDPM

- A **U-Net network** is trained to predict the facial region where a mask should be placed from a clean face image.
- The predicted **mask region is combined with the clean face** to define the editable area
- A **diffusion model trained from scratch** generates the masked region through iterative denoising



Stable Diffusion Inpainting

- Uses pretrained Stable diffusion inpainting to generate the masked region
- **LoRA fine-tuning** adapts the model to produce realistic medical-style masks



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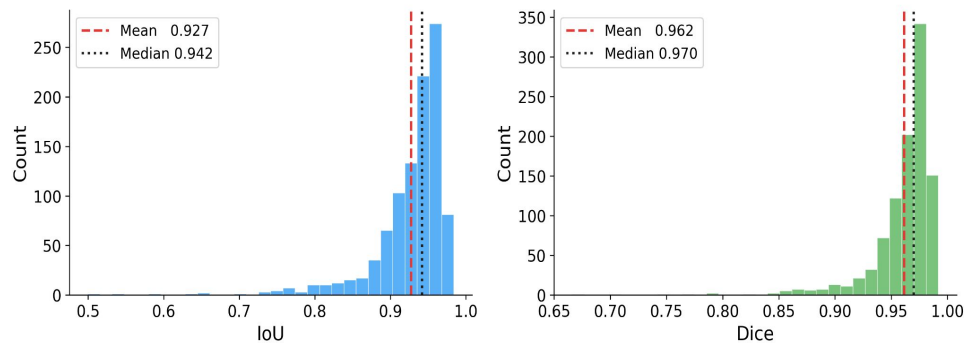
Experimental results

We finetuned the models using 2k data for training , 1k data for testing

Results on Conditional-Unet mask predictor - 30 epochs



P2 — IoU & Dice Distribution over Test Set



Training Procedure	Train FID	Test FID	Epoch
DDPM	47.087	48.93	50
CycleGAN	28.43	28.48	30
SD-Inpainting	36.54	32.79	8

CycleGAN



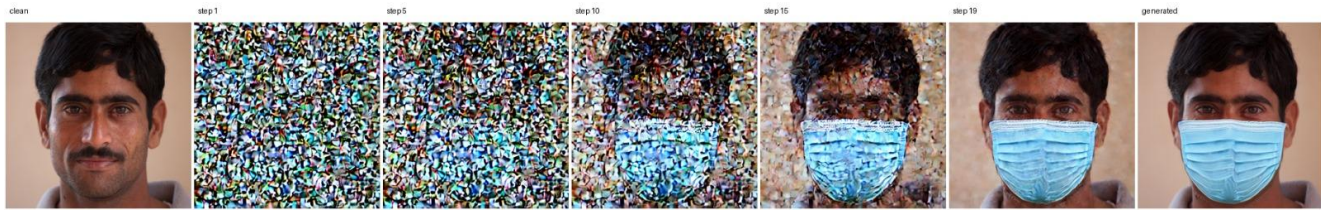
DDPM



SD-Fine tuning



Denoising process of SD-Inpainting- 20 steps



Failure case

Our attempt using Stable Diffusion inpainting model underperforms the baseline on FID. This may be due to several factors, including noisy mask predictions from the U-Net, limited hyperparameter tuning. Because the model relies on accurate binary masks, small localization errors can introduce artifacts and negatively affect FID score.

Discussion

- At inference time, CycleGAN may produce good masked-face results only if the test image is sufficiently similar to the training distribution. In particular, if the test image is not centered, not aligned, captured under an extreme pose, or of low quality, performance may degrade
- Our method is more flexible because, at inference time, it uses both clean face and a binary mask to indicate where the facial mask should be generated. As a result, it is more applicable to images from other distributions

References

- [1] S. Mumford, Generation of Realistic Facemasked Faces With GANs, 2021
- [2] R. Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, CVPR, 2022
- [3] E. Hu et al., LoRA: Low-Rank Adaptation of Large Models, 2021
- [4] T. Karras et al., A Style-Based Generator Architecture for Generative Adversarial Networks, CVPR, 2019
- [5] A. Cabani et al., MaskedFace-Net: A Dataset of Masked Faces, 2021